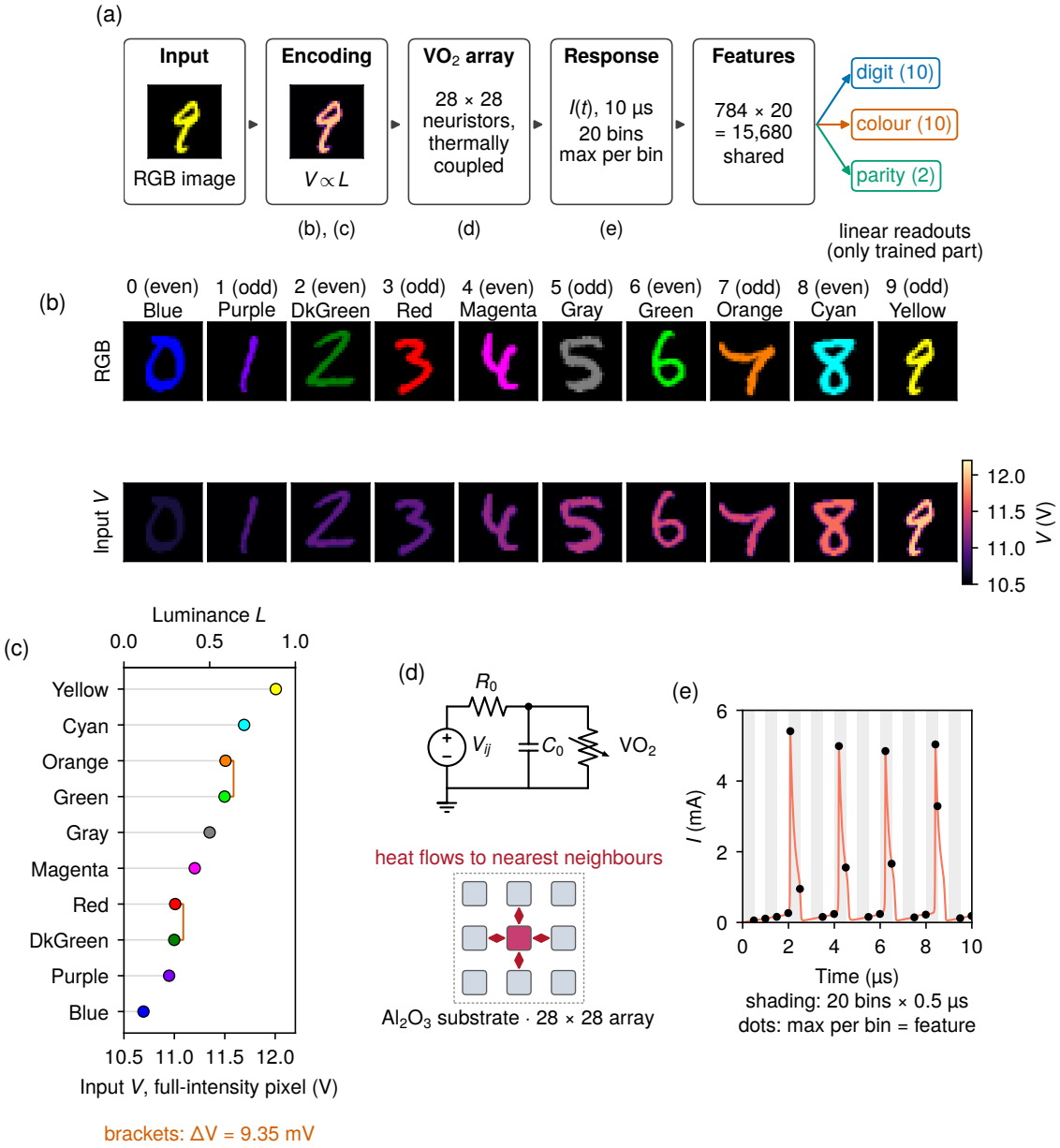

(a)
Input
RGB image
Encoding
$V \propto L$
$VO_2$ array
28 × 28 neuristors, thermally coupled
Response
$I(t)$, 10 µs
20 bins
max per bin
Features
784 × 20 = 15,680 shared
digit (10)
colour (10)
parity (2)
(b), (c)
(d)
(e)
linear readouts (only trained part)
(b)
0 (even) Blue
1 (odd) Purple
2 (even) DkGreen
3 (odd) Red
4 (even) Magenta
5 (odd) Gray
6 (even) Green
7 (odd) Orange
8 (even) Cyan
9 (odd) Yellow
RGB
Input $V$
12.0
11.5
11.0
10.5
$V$ (V)
(c)
Luminance $L$
0.0
0.5
1.0
Yellow
Cyan
Orange
Green
Gray
Magenta
Red
DkGreen
Purple
Blue
10.5
11.0
11.5
12.0
Input $V$, full-intensity pixel (V)
brackets: ΔV = 9.35 mV
(d)
$R_0$
$V_{ij}$
$C_0$
$VO_2$
heat flows to nearest neighbours
$Al_2O_3$ substrate · 28 × 28 array
(e)
$I$ (mA)
0
2
4
6
0
2
4
6
8
10
Time (µs)
shading: 20 bins × 0.5 µs
dots: max per bin = feature